Kyle Genova

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EDUCATION

Sept. 2016 - May 2021 **Princeton University** Princeton, NJ

- **Ph.D. Computer Science**
- **Advisor: Prof. Thomas Funkhouser**
- **2021 Siebel Scholar**
- **National Science Foundation GRFP Fellowship**
- **Gordon Y.S. Wu Fellowship in Engineering**

Aug. 2012 - May 2016 **Cornell University College of Arts and Sciences** Ithaca, NY

- **B.A. Computer Science**
- **GPA: 4.17**
- **Phi Beta Kappa (Top 3% of class)**

EXPERIENCE

May 2024 - Present **Senior Research Scientist at Google DeepMind** New York, NY

- Researcher: Foundational Research Unit
- ***Training Multimodal Models (3D , 2D, and Language) for Large-Scale 3D Feature Learning***
- ***Gemini Large-Scale Pretraining Contributor***

May 2021 - May 2024 **Research Scientist at Google Research** Mountain View and NYC

- Publishing papers, collaborating with product teams.
- ***30 million updates to Google Maps based on 3D learning***
- ***Published 3D learning method (2D3DNet) that launched as a Google Maps service***

May. 2016 - April 2021 **5 Internships at Google Research** NYC, California, Boston

- Multiple research projects in 3D Vision, Graphics, and Machine Learning
- ***9 Publications + 4 Patents***

Sept. 2017 - May. 2018 **Teaching Assistant at Princeton University** Princeton, NJ

- Computer Vision (COS 429), Computer Graphics (COS 426)
- ***Graduate Student Teaching Award***

PUBLICATIONS (4,528 Citations, 20 Papers, 5 Patents, 1 CVPR Best Paper Nom.)

Visual Chronicles: Using Multimodal LLMs to Analyze Massive Collections of Images. Boyang Deng, Songyou Peng, Kyle Genova, Gordon Wetzstein, Noah Snavely, Leonidas Guibas, Thomas Funkhouser. In submission. 2025.

LOC3DIFF: Local Diffusion for 3D Human Head Synthesis and Editing. Yushi Lan, Feiton Tan, Qiangen Xu, Di Qiu, Kyle Genova, Zeng Huang, Sean Fanello, Rohit Pandey, Thomas Funkhouser, Chen Change Loy, Yinda Zhang. ECCV 2024

NIFTY: Neural Object Interaction Fields for Guided Human Motion Synthesis. Nilesh Kulkarni, Davis Rempe, Kyle Genova, Abhijit Kundu, Justin Johnson, David Fouey, Leonidas Guibas. CVPR 2024

OpenScene: 3D Scene Understanding With Open Vocabularies. Songyou Peng, Kyle Genova, Chiyu Jiang, Andrea Tagliasacchi, Marc Pollefeys, Thomas Funkhouser. CVPR 2023

Nerflets: Local Radiance Fields for Efficient Structure-Aware 3D Scene Representation from 2D Supervision. Xiaoshuai Zhang, Abhijit Kundu, Thomas Funkhouser, Leonidas Guibas, Hao Su, Kyle Genova. CVPR 2023

Polynomial Neural Fields for Subband Decomposition and Manipulation. Guandao Yang, Sagie Benaim, Varun Jampani, Kyle Genova, Jonathan Barron, Thomas Funkhouser, Bharath Hariharan, Serge Belongie. NeurIPS 2022

Panoptic Neural Fields: A Semantic Object-Aware Neural Scene Representation. Abhijit Kundu, Kyle Genova, Xiaoqi Yin, Alireza Fathi, Caroline Pantofaru, Leonidas J. Guibas, Andrea Tagliasacchi, Frank Dellaert, Thomas Funkhouser. CVPR 2022

Differentiable Surface Rendering via Non-Differentiable Sampling. Forrester Cole, Kyle Genova, Avneesh Sud, Daniel Vlasic, Zhoutong Zhang. ICCV 2021

Multi-Resolution Deep Implicit Functions for 3D Shape Representation. Zhang Chen, Yinda Zhang, Kyle Genova, Sean Fanello, Sofien Bouaziz, Christian Häne, Ruofei Du, Cem Keskin, Thomas Funkhouser, Danhang Tang. ICCV 2021

IBRNet: Learning Multi-View Image-Based Rendering. Qianqian Wang, Zhicheng Wang, Kyle Genova, Pratul P. Srinivasan, Howard Zhou, Jonathan T. Barron, Ricardo Martin-Brualla, Noah Snavely, Thomas Funkhouser. CVPR 2021

NESF: Neural Semantic Fields for Generalizable Semantic Segmentation of 3D Scenes. Suhani Vora, Noha Radwan, Klaus Greff, Henning Meyer, Kyle Genova, Mehdi S.M. Sajjadi, Etienne Pot, Andrea Tagliasacchi, Daniel Duckworth. CVPR 2021

Learning 3D Semantic Segmentation with only 2D Image Supervision. Kyle Genova, Xiaoqi Yin, Abhijit Kundu, Caroline Pantofaru, Forrester Cole, Avneesh Sud, Brian Brewington, Brian Shucker, Thomas Funkhouser. CVPR 2021

Local Deep Implicit Functions for 3D Shape. Kyle Genova, Forrester Cole, Avneesh Sud, Aaron Sarna, Thomas Funkhouser. CVPR 2020 ***(Oral)***.

CvxNet: Learnable Convex Decomposition. Boyang Deng, Kyle Genova, Soroosh Yazdani, Sofien Bouaziz, Geoffrey Hinton, Andrea Tagliasacchi. CVPR 2020 **(*Best Paper Nominee)***.

Towards Fairness in Visual Recognition: Effective Strategies for Bias Mitigation. Zeyu Wang, Klint Qinami, Ioannis Christos Karakozis, Kyle Genova, Prem Nair, Kenji Hata, Olga Russakovsky. CVPR 2020.

Learning Shape Templates with Structured Implicit Functions. Kyle Genova, Forrester Cole, Daniel Vlasic, Aaron Sarna, William T. Freeman, Thomas Funkhouser. ICCV 2019.

Text-based Editing of Talking-head Video. Ohad Fried, Ayush Tewari, Michael Zollhöfer, Adam Finkelstein, Eli Schechtman, Dan B. Goldman, Kyle Genova, Zeyu Jin, Christian Theobalt, Maneesh Agrawala. SIGGRAPH 2019.

Unsupervised Training for 3D Morphable Model Regression. Kyle Genova, Forrester Cole, Aaron Maschinot, Aaron Sarna, Daniel Vlasic, William T. Freeman. CVPR 2018 **(*Spotlight)***.

Learning Where to Look: Data-Driven Viewpoint Set Selection for 3D Scenes. Kyle Genova, Manolis Savva, Angel X. Chang, Thomas Funkhouser. CoRR 2017.

An Experimental Evaluation of the Best-of-Many Christofides' Algorithm for the Traveling Salesman Problem. Kyle Genova & David P. Williamson. Algorithmica 2017 ***(ESA Top Papers)***, ESA 2015.

CONFERENCES

ICCV 2025: Area Chair

SKILLS

- Python, C++, CUDA, JAX, TensorFlow, PyTorch, OpenGL, GLSL